

Block

4

ANALYSIS & INTERPRETATION

UNIT 13

Basic Statistical Analysis

UNIT 14

Data Analysis

UNIT 15

Report Writing



UNIT 13 BASIC STATISTICAL ANALYSIS

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13.0 INTRODUCTION

Numbers are part of our daily life; we are used to coming across various sets of numbers meaning something to us. For example, a mild earthquake measured 4 Richter scale hits a mountain region but a severe 7 point Richter scale earthquake killed 100s of people. The difference between 4 and 7 numbers is just 3, but the difference in terms of magnitude - the former is a mild one and later is a destructive force. So, numbers have meaning, when comes to complex sets of data, we need a scientific tool to classify and understand these data and make sense of them. The process to understand data is data analysis and the tool to do this is called statistics. In Communication Studies, statistical analysis is generally employed in quantitative settings, and it has a limited scope in qualitative research.

In this unit, we shall discuss different tools of descriptive statistical analysis in detail with the help of exercises.

13.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- discuss the need and importance of statistical analysis in communication research;
- describe Populations, Samples and Scales of measurement;
- explain Frequency Distribution;
- define Variability;
- discuss Standard Deviation and Variance for a population;
- explain measures of Central Tendency such as mean, mode and median; and
- use above statistical methods while analysing your data

13.2 INTRODUCTION TO STATISTICS

Statistics is a branch of mathematical sciences which deals with numbers and data sets. A basic definition of statistics is, “it is a process and tool involved in collection, analysis, interpretation, explanation and presentation of data.” Through various tools of statistical analysis, we can extract information from raw data, and create a new understanding from a set of numbers. Statistics help us to simplify complex data sets. Statistical tools help us to interpret and explain information. It helps us to make informed decisions and even helps us to predict a direction based on diverse data sets.

Statistics are broadly classified into two types: descriptive statistics and inferential statistics. Descriptive statistics, as the name suggests, is meant to classify and summarise collected raw data. Major kinds of analysis under descriptive statistics are: measures of frequency, measures of central tendency and measures of dispersion. The second type, inferential statistics help us to draw conclusions from association or differences between variables.

As this unit deals with the basic concepts of statistical applications, we shall focus on various analytical tools of descriptive statistics in detail along with the help of exercises.

13.2.1 Populations and Samples

Block 1, Unit 4 is devoted to sampling. You could go back to that unit to refresh yourself about sampling. Here, just to recap a population is the entire group of individuals that a researcher wishes to study. A sample is a subset of a population. It is drawn from out of the population as defined by the researcher according to the rules of sampling.

13.2.2 Scales of Measurement

Measurement is the heart and soul of research. One of the central characteristics of a quantitative study is the requirement to measure whatever you study: gender, occupation, height, weight, time, viewership, motivation, care, commitment, gratification, loyalty... etc. When measuring, we assign numbers (1, 2, 3, etc) to variables and concepts under investigation. For example, if you have a variables

such as gender with two categories (female and male), you will measure it quantitatively by assigning a unique number to each of the two categories of gender (you may assign number 1 to the female category and 2 to the male category).

Numbers can also be placed into a scale based on their inherent characteristics such as the kind of number. We will briefly look at numbers in terms of scale.

Nominal Scale

This is the lowest scale (or level) of measurement. At this level, each response category of a variable is assigned a unique number. For example, as discussed above, a variable such as gender having two response categories will assign a unique or distinct number to each of the two categories. At a nominal level, the category is used simply to describe.

No mathematical relationships are possible at the nominal scale of measurement. You cannot add, subtract, multiply or divide. You can only count

Ordinal Scale

The word 'ordinal' is derived from the word, 'order'. In an ordinal scale of measurement, numbers indicate only the rank order of cases (respondents) on some variable.

For example, all students **who** pass an examination get I, II or III class. In this case, you will be able to say that a student who earned an I class has fared better in the examination than a student who has secured a II class; similarly, a student who passed in II class has fared better than a student who passed in III class. That is, you will be able to rank students in the order of their results. But you will not be able say the distance (interval) between any two ranks.

Interval Scale

Interval measurement has the properties of the nominal and ordinals scales, plus the quality that equal interval between numerals indicate equal distance in the variables being measured. For example, if the temperature today is 20°C and on May 1 it was 40°, then you can say that the temperature today is 20°C lower than on May 1. In other words with an interval scale you can determine precisely which observation is of greater magnitude and by how much it is greater. You are able to do so by subtracting one number from the other. This tells us something important about interval scales: measurements on an interval scale permit basic arithmetic operations such as subtraction and addition.

However, measurements on an interval scale do not allow you to multiply or divide. Let us explain this with the example of temperature: Assume the temperature to be 20°C today and the temperature to be 40° C on a given day in summer. The only statement you can make in this case is that the difference between the temperatures on the two given dates is 20° C. But you cannot say that it is half as hot today as on a given day in summer. Why is it so? Because a ratio (20° C: 40° C) is meaningful only when the scale of measurement has an absolute zero point that indicates the total absence of the concept being measured. For example, zero degree on the Centigrade scale does not mean absence of temperature. This point becomes clearer if you consider both temperatures measured on the Celsius as well as the Fahrenheit scales.

Interval scales consist of numbers that rank items such that numerically equal distances on the scale represent equal distances in the property being measured. However, the 0 on the scale is arbitrary; that is, 0 does not have a true or absolute value and it does not indicate the absence of property measured.

Ratio scale

The fourth level, called ratio measurement, retains the properties of the three preceding scales plus has additional property of an absolute zero point. Because a ratio scale of measurement has an absolute zero point, ratios of numbers on this scale reflect ratios of magnitudes. For example, the variable income, measured in rupees, has this property. If one person's annual income is 10 lakhs and another's person annual income is 20 lakhs, you can divide one with the other (that is form a ratio) to state that one is twice (or half) as much as the other.

Ratio scales consist of numbers that rank items such that numerically equal distances on the scale represent equal distances in the property being measured. Importantly, the 0 on the scale means absence of property measured. The 0 therefore has real meaning.

13.2.3 Discrete and Continuous Variables

People who design research pay close attention to classification of variables. Although one can classify variables in different ways, we will examine two types of variables here: discrete and continuous. Let us do so by looking at some questions first:

- How many siblings do you have?
- How many newspapers are published from your city? What is your marital status?

The answer to each of the three questions is a whole number. A discrete variable consists of separate, indivisible categories. It cannot be divided into fractional parts. You can have two or three siblings, but you can't have 2.5 siblings, for example.

A discrete variable has no intermediate values between two adjacent categories.

Now examine another set of questions you generally read in a questionnaire. What is your monthly household income?

- What is your age?

Response to the two questions need not necessarily be whole numbers. Your income can be measured in rupees and paise, your age can be measured in years, months, days, even minutes. So the two variables in this instance (monthly household income (MHI) and age) are not discrete. Variables such as MHI, age, time, and weight are not limited to a fixed set of separate, indivisible categories. These variables are called continuous because they can be divided into an infinite number of fractional parts.

A continuous variable is divisible into an infinite number of fractional parts.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

- 1) Employment status: working full-time, working part-time, student, housewife, unemployed.

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- 2) Religion: Christian, Muslim, Sikh, Jain, Hindu, other.

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- 3) Gravity of crime: most serious, moderately serious, least serious.

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- 4) Agreement or disagreement: asking respondents to state their level of agreement on a scale such as strongly agree, somewhat agree, neutral, somewhat disagree, strongly disagree.

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- 5) Number of items purchased: Shoppers in a mall indicate the number of items purchased in the mall.

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13.3 FREQUENCY DISTRIBUTION

When you work with a small dataset, it is easy to make sense of it. But when you are dealing with a large dataset (let's say more than 100 respondents have

answered 70 questions on a questionnaire), then it is really difficult to make sense of the data or see any pattern or trend in it. This is where frequency distribution helps. Simply put, a frequency distribution organises the data either as a table or graph making it easier to see a clear pattern or trend in it. One of the most common procedures for organising a set of data is to place the scores in a frequency distribution.

A frequency distribution is a record of the number of individuals located in each category on the scale of measurement. It allows the researcher to see “at a glance” the entire set of scores.

13.3.1 Frequency Distribution Tables

The simplest frequency distribution table presents the list of the individual scores in a column from highest to lowest. Besides each score, in another column, is the frequency, or number of times the scores occurred in the data. It is customary to use an X as the column heading for the scores and an f as the column heading for the frequencies.

Example: A class of 20 students scores the following marks out of 10 ($N = 20$ scores).

We will organise these scores by constructing a frequency distribution table.

- First, identify the highest score in the set of scores. In this set, the highest score (X) is 10.
- Next, identify the lowest score in the set of score. In the set, it is (X) 4.
- Write down the scores from the highest to the lowest (10 to 4) in a column, irrespective of whether a score appears in the set of scores or not (5 doesn't appear in the set of scores but it is identified, as shown in the table). This is the first column in the table.
- At the fourth step, against each number, count and write down the frequency (f) with which it appears. For example, 10 appears twice, 9 appears 7 times, so on and so forth. This is the second column of the table.
- Finally, in the third column, multiply each X value by its frequency and then add these products. This gets the “ $\sum fX$ ” from a frequency distribution table. This sum may be expressed in symbols as “ $\sum fX$ ”. In addition, by adding up the frequencies, you will obtain the total number of individuals: “ $\sum f = N$ ” (20 students or scores in the example below)

Score	Frequency	Sum
X	f	fX
10	2	20
9	7	63
8	5	40
7	2	14
6	3	18
5	0	0
4	1	4
	20	$\sum fX = 159$

The concept of a continuous variable and real limits are useful in measurement. Let us therefore try to understand them with a simple example. How do you measure time and the number of subjects you study in any degree programme? You could measure time in years or even break it down into months, days, hours, minutes, seconds, even nanoseconds, if you have sophisticated instruments. But you cannot do the same thing with the number of subjects you study in your degree programme. In the above example, time is a continuous variable. A continuous variable has an infinite number of possible values. It has no gaps in its scale (that is why, continuous). Therefore it may be represented by a number line that is continuous and contains an infinite number of points (see below).

Let's take another example. When you measure body weight to the nearest kilogram, weights of $X = 50.3$ and $X = 49.6$ kgs would both be rounded to scores of $X = 50$. Note that a score of $X = 50$ is an approximate number; it is not a single point on the number line. The actual body weight is somewhere between 49.5 kgs and 50.5 kgs. Any measurement that falls within this interval will be assigned a value of $X = 50$. The boundaries that form the interval are called the real limits of the interval that corresponds to $X = 50$.

X	F
X	Y
12 - 13	4
10 - 11	5
8-9	3
6-7	3
4-5	2



Real Limits

Real limits are the boundaries of intervals for scores that are represented on a continuous number line. The real limits separating two adjacent scores is located exactly between the scores. Each score has two real limits. The upper real limit is at top and the lower real limit is at the bottom of the interval. The concept of real limits also applies to the class intervals of a grouped frequency distribution table. For example, a class interval of 40 - 49 contains scores from $X = 40$ to $X = 49$. These values are called the apparent limits of the interval because it appears that they form the upper and lower boundaries for the class interval. But $X = 40$ is actually an interval from 39.5 to 40.5. Similarly, $X = 49$ is an interval from 48.5 to 49.5. Therefore, the real limits of the interval are 39.5 (the lower real limit) and 49.5 (the upper real limit).

13.3.3 Frequency Distribution Graphs

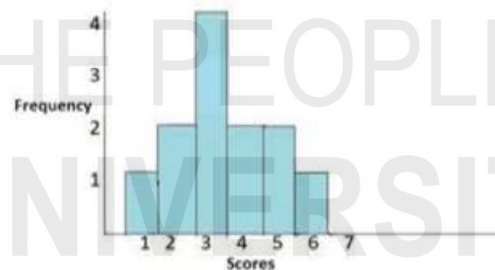
A frequency distribution graph is a pictorial representation of the data shown in a frequency distribution table. All the graphs start with perpendicular lines called axes. The horizontal line is called the X - axis, and the vertical line is called the Y - axis. The scores are listed along the X - axis in increasing value from left to right. The frequencies are listed on the Y - axis in increasing value from bottom to top. As a general rule, the point where the two axes intersect should have a value of zero for both the scores and the frequencies. A final general rule is that the graph should be constructed so that its height (Y - axis) is approximately three quarters of its length (X - axis).

13.3.4 Histograms and Bar Graphs

For this type of graph, you simply draw a bar above each score so that the height of the bar corresponds to the frequency of the score. The choice between using histogram or a bar graph is determined by the scale of measurement. When a frequency distribution graph is showing data from an interval or ratio scale, the bars are drawn so that adjacent bars touch each other. The touching bars produce a continuous figure which emphasises the continuity of the variable. This type of frequency distribution graph is called a histogram. For a histogram, vertical bars are drawn above each score so that

- 1) The height of the bar corresponds to the frequency
- 2) The width of the bar extends to the real limits of the score.

<i>X</i>	<i>F</i>
6	1
5	2
4	2
3	4
2	2
1	1



E.g.: Frequency Distribution Histogram.

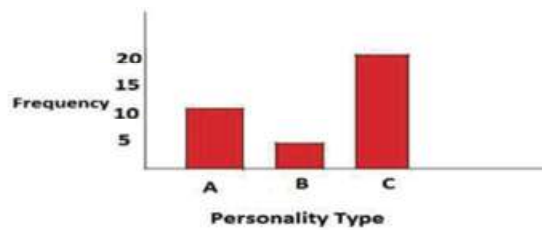


E.g.: Frequency Distribution Histogram for grouped data.

When you are presenting the frequency distribution for data from a nominal or ordinal scale, the graph is constructed so that there is some space between the bars. In this case the separate bars emphasise that the scale consists of separate, distinct categories. The resulting graph is called a bar graph. For a bar graph, a vertical bar is drawn above each score (or category) so that

- 1) The height of the bar corresponds to the frequency
- 2) There is a space separating each bar from the text.

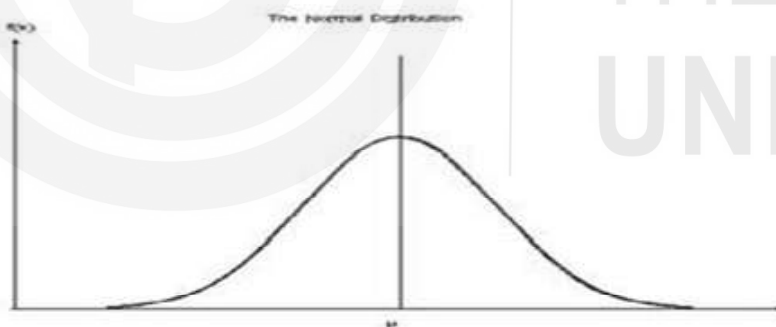
E.g.: Bar Graph showing the distribution of personality types in a sample of college students. Because personality type is a discrete variable measured on a nominal scale, the graph is drawn with space between the bars.



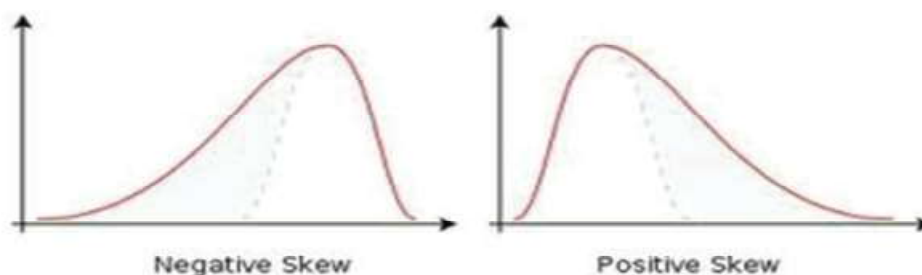
13.3.5 Shape of a Frequency Distribution

There are three characteristics that completely describe any distribution: Shape, central tendency and variability. We will later examine central tendency and variability. Let's try understanding **now** the shape of the distribution. Nearly all distributions can be classified as being either symmetrical or skewed.

- In a symmetrical distribution it is possible to draw a vertical line through the middle so that one side of the distribution is an exact mirror image of the other.
- In a skewed distribution the scores tend to pile up towards one end of the scale and taper off gradually at the other end. The section where the scores taper off towards one end of a distribution is called the tail of the distribution. A skewed distribution with the tail to the right hand side is said to be positive skewed because the tail points toward the positive (above zero) end of the X-axis. If the tail points to the left, the distribution is said to be negatively skewed.



Symmetrical Distribution



Positive and Negative Skew

Check Your Progress 2

Find frequency distribution for Male journalists, Female journalists and Total Journalists.

S.No.	Age Intervals	Male	Female
1	Under20	0	100
2	21-25	17	83
3	26-30	40	60
4	31-35	35	65
5	36-40	51	49
6	41-45	48	52
7	46-50	55	45
8	51-55	56	44
9	56-60	53	47
10	61-65	56	44
11	Over 65	67	33

Find frequency distribution for Girls, Boys and Total Class. Dataset is about the time spent by a class of students in watching television programmes in a month.

S.No.	Time Intervals	Girls	Boys
1	140-149	0	0
2	130-139	0	32
3	120-129	0	48
4	110-119	1	29
5	100-109	0	18
6	90-99	3	14
7	80-89	5	5
8	70-79	6	5
9	60-69	14	0
10	50-59	7	1
11	40-49	11	0
12	30-39	4	0

13.4 MEASURES OF CENTRAL TENDENCY

You are quite familiar with what an average is and how it is calculated. When asked to calculate the average most of us will rush into calculating the mean. But this isn't exactly the only meaning of the 'average'. In statistics, you can calculate the so-called average in three different ways. Collectively, these three - mean, medium, and mode, are called measures of central tendency. You can pretty much guess what central tendency is. As the phrase connotes, central tendency is an

average that resides in the centre or represents the centre of the distribution. Measures of central tendency give us a number that describes where the center lies. Knowing the centre of a distribution tells us a great deal about a distribution, because most of the scores in a distribution tend to cluster around the centre.

13.4.1 Mean

The mean is the arithmetic centre of the distribution. You can find the mean by adding all the scores (OX) together and dividing the total by the number of scores (N). For example (X = scores 1, 2, 3, 4, 5; N=5):

$$\begin{aligned} \text{Xare } 1+2+3+4+5 &= (\text{OX}) 15 \\ \text{N} &= 5 \end{aligned}$$

The population mean (μ) is: $15/5 = 3$ ($\mu = \text{OX}/\text{NJ}$). The sample mean (n) is: $15/5 = 3$ ($x = \text{OX}/n$)

You may now note that the procedure we used to calculate the mean of the population and the sample is the same, but we have used different symbols for them.

We now know how the mean is calculated. Means have four properties. Here are four of them:

- 1) The mean is the most reliable and most often used measure of central tendency.
- 2) The mean need not be an actual score in the distribution.
- 3) The mean is strongly influenced by outliers.
- 4) The sum of the deviations about the mean must always be equal to zero.

You should understand the last two properties of the mean listed above. First, you should know that an outlier is an extreme score. It is a score that stands apart from most of the scores in the distribution. Thus, in a set of scores 1, 2, 9, the score 9 stands apart from the rest of the scores. 9 is an outlier.

Outliers tend to pull the mean towards them. Thus a distribution of values 1, 2, 3, 4, 5 has an average of 3. Three does a good job of describing where most of the scores in this distribution lie. However, if there is an outlier, say 25 (instead of 5) in the above distribution, then the mean changes a great deal. The new distribution 1, 2, 3, 4, 25 has a mean of 7. Seven is not really close to most of the other values in the distribution. Thus, the mean is a poor measure of the centre when we have outliers.

13.4.2 Median

The median is the physical centre of the distribution. It is the value in the middle when the values of the distribution are arranged sequentially. The distribution: 0, 1, 2, 2, 3, 4, 5, 5, 5, 7, 8 has a median value of 4 because there are five values above this point and five values below this point in the distribution (0, 1, 2, 2, 3, 4, 5, 5, 5, 7, 8).

If you have an even set of numbers then there will be two values that are at the centre, and you average these two values together in order to determine the median. For example, if we take out one of the numbers in the distribution so that we have 0, 1, 2, 2, 3, 4, 5, 5, 5, 7 then the two values in the centre are 3 and 4 (0, 1, 2, 2, 3, 4, 5, 5, 5, 7). The average is 3.5 and that is the median. You would

have noticed that the median is resistant to outliers. That is, outliers will generally not affect the median and it will not be affected as much as the mean. It is possible the median might move slightly in the direction of the skew or outliers in the distribution.

13.4.3 Mode

The mode is the most frequent score in the distribution. Examine the following scores. 1, 1, 2, 3, 3, 3, 4, 4, 4, 4, 5, 6, 7. What score appears most often in the distribution? Obviously 4. Hence, 4 is the mode. This distribution of scores has only one mode (4).

Now examine another set of scores: 1, 1, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 6. Here 3 and 4 appear four times each. There are, therefore, two modes (3 and 4) in the set of scores.

Now how about this set of scores: 1, 2, 3, 4, 5, 6, 7, 8, 9. Here each score appears only once. No score appears twice or more. You, therefore, conclude there is no mode in the set of scores. You would have again noticed another characteristic of mode: it remains unaffected by outliers.

- 1) In a class, students were asked to select their favourite subject. 63 students chose English, 29 students chose Maths and 9 students chose statistics. What is the mode of this distribution?
- 2) Why is the median not affected by extreme scores (outliers)?

Check Your Progress 3

Find the mode, median, mean of the following:

- a) 19,24,28,28,30,34,38,42,49,51,52,61,64

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- b) 53,43, 77,44,35, 14,56,43, 75,45,80

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- c) 14,25,47,56,56,58,58,66,82,89,91

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- d) 61,52,81,56,35,17,56,51,82,35

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- e) 31,44,45,46,43,55,56,56,57,62,64,66,69

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13.5 VARIABILITY

Our education and experience enable us to understand what a mean is. However, understanding the concept of variability doesn't come to us so easily.

Variability, in simple English, refers to difference (or distance). So, if a population of 10 students scores 10 out of 10, it certainly is great performance, but there is no difference in the score of the 10 students. They have all scored exactly the same marks! The statistician would say the same thing in a different language (there is no variability in the distribution of (10) scores, she would say). On the other hand, if the 10 students score 1 through 10, then there would be a great amount of variability.

In statistics our goal is to measure the amount of variability for a particular set of scores. If the scores in a distribution are all the same, then there is no variability. If differences are small, then the variability is small, and if differences between scores are large, then the variability is large.

Now, in our everyday language, 'small', and 'big' are qualitative expressions. Each one of has our own subjective measure of what these two words mean. In statistics though, variability is quantitatively measured and expressed.

Why do we measure variability? We measure variability in order to determine how spread out a distribution of scores is. The purpose is to find out whether the scores are all close to each other (example: 2, 4, 4, 4, 5, 5) or are they scattered over a wide range of values (0, 0, 1, 2, 6, 9). Variability does much more than describing a set of scores. It helps us understand two additional characteristics of the distribution. First, variability tells us how well the mean (average) describes the distribution. If the variability is small, then the scores are all close together, and each individual score is close to the mean (calculate the mean of the first set of scores and see this for yourself). In this situation, the mean is a good representative of all the scores in the distribution. On the other hand, when variability is large, the scores are all spread out, and they are not necessarily close to the mean (calculate the mean of the second set of scores) Is the mean of the second set of scores as good a representative of the distribution as the mean of the first set?

The second characteristic of variability is kind of related to the first. Variability also tells us how well an individual score (or group of scores) represents the entire distribution. In the first set of scores (2, 4, 4, 4, 5, 5) the distance of any individual score (2 or 4 or 5) is not too far from the rest of the scores. In this case, 2 or 4 or 5 nearly accurately represent the entire distribution. Can you say the same thing about the second distribution? Does any individual score represent the entire distribution? We can't be so certain here or as certain as we are with the first set of scores. Now with this little theoretical background, let's try to find out how variability is calculated. There are three different measures of variability: the range, the interquartile range and the standard deviation. Let's look at each of them.

13.5.1 Range

The range is the distance (or difference) between the largest score and the smallest score in the distribution. So, the range of the two sets of scores given above is 3

and 9 respectively. (To be accurate the range is the difference between the upper real limit for the largest (maximum) X value and the lower real limit of the smallest (minimum) X value).

You can perhaps see the difficulty in using range as a measure of variability. It seeks inputs from only two extreme values (scores) at the cost all other values in the distribution. In other words, it often does not give an accurate description of the variability for the entire distribution. For this reason, range is considered to be a crude and unreliable measure of variability.

13.5.2 Interquartile Range and Semi-Interquartile Range

The second measure of variability is the interquartile range (or semi-quartile range). You can divide a distribution into four equal parts using quartiles (for easier understanding think of dividing 100 into four equal parts!).

The first quartile (Q1) is the score that separates the lower 25% of the distribution from the rest 75 per cent.

The second quartile (Q2) is the score that has exactly two quartiles, or 50% of the distribution below it. The second quartile is the median (they are same).

Finally the third quartile (Q3) is the score that divides the bottom 75 per cent of the distribution from the top 25 per cent.

The semi quartile range is one half of the Interquartile range. It provides a descriptive measure of the “typical” distance of scores from the median (Q2).

The Interquartile range is the distance between the first quartile and the third quartile.

Interquartile range = $Q3 - Q1$

When the Interquartile range is used to describe variability, it commonly is transformed into the semi-interquartile range. The semi-interquartile range is simply one half of the Interquartile range.

$$\text{Semi-interquartile range} = \frac{(Q3 - Q1)}{2}$$

Because the semi Interquartile range focuses on the middle 50% of the distribution, it is less likely to be influenced by extreme scores and, therefore gives a more stable measure of variability than the range. However, much like the range, the semi-interquartile range does not take into account the actual distances between individual scores. In other words, it doesn't tell us so it how scattered or clustered the scores are.

E.g.: Q2 (the middle quartile) is the median.

Q1 (the lower quartile) is the median of the numbers to the left of, or below Q2.
Q3 (the upper quartile) is the median of the numbers to the right of, or above Q2.
Example: Find the lower, middle and upper quartiles of the data

12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32

Since the data is already in ascending order, find the median. 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32

22 is the median, therefore, $Q_2 = 22$

The median of the numbers to the left of Q_2 : 12, 14, 16, 18, 20 16 is the median, therefore, $Q_1 = 16$

The median of the numbers to the right of Q_2 : 24, 26, 28, 30, 32 28 is the median, therefore, $Q_3 = 28$

The inter quartile range of a distribution is the difference between the upper and lower quartiles.

That is, inter quartile range = $Q_3 - Q_1$

Since, $Q_3 = 28$ and $Q_1 = 16$

Inter quartile range (Q_2) = $28 - 16 = 12$

The semi-interquartile range of a distribution is half the difference between the upper and lower quartiles, or half the inter quartile range.

Therefore, from the example above, the inter quartile range = 12.

Therefore, Semi Interquartile range = $\frac{(Q_3 - Q_1)}{2} = \frac{12}{2} = 6$

13.5.3 Standard Deviation and Variance for a Population

Now we come to the third measure of variability: the standard deviation. Unlike the first two measures of variability, the standard deviation takes into account the actual distances between individual scores. So, naturally, it tells us something important that the two other previous measures don't: how scattered or clustered the scores are in a distribution.

Well, you must, by now, be wondering, how does one calculate the actual distance between individual scores? Think of it this way. How is distance measured in a city? The standard manner of measuring distance is from a central point to any other point. The same logic is used in calculating the standard deviation. The central point, in this instance, is the mean. So, we measure the deviation of the scores from the mean.

Standard means typical; deviation is distance from the mean

The standard deviation can be measured in six simple steps. Let us illustrate this using a set of scores (2, 4, 4, 4, 5, 5).

Step 1 is to add up the scores. $2 + 4 + 4 + 4 + 5 + 5 = 24$

Step 2 is to compute the mean of the set of scores. You do this by dividing the total (24) with the number of scores ($N = 6$)

$24 \div 6 = 4$

Step 3 is to calculate the mean of the deviation scores. To do this, you subtract the mean from each of the scores and add up the deviation scores.

$$\begin{array}{r}
 2-4 = -2 \\
 4-4 = 0 \\
 4-4 = 0 \\
 4-4 = 0 \\
 5-4 = 1 \\
 5-4 = 1 \\
 \hline
 0
 \end{array}$$

When you do this you end up with zero (because -2 and + 2 cancel each other out). You will observe that the distances above the mean are equal to the distances below the mean. This will always be so. Therefore, the mean deviation score provides no information about variability.

Step 4 allows you to get over this by squaring the difference between the mean and individual scores (squaring helps get rid of the + and - signs) and adding them up. When you do this for the set of scores in the example, you get.

$$2-4 = -2 = 4$$

$$4-4 = 0 = 0$$

$$4-4 = 0 = 0$$

$$4-4 = 0 = 0$$

$$5-4 = 1 = 1$$

$$5-4 = 1 = 1$$

$$6$$

At this step you not only have got rid of the signs, you also are able to add up the squared deviation scores and get a value that is not zero (in this case, 6). Several textbooks call this Sum of Squares or SS. Sum of Squares is an important concept and is used in several statistical tests. You must remember what it is and how it is calculated.

Sum of squares (SS) is the sum of the squared deviation scores.

Step 5 takes us to another important concept called variance. You compute variance by dividing the sum of squares by N. In the example,

$6 / 6 = 1$. This is the variance

Variance is the mean of the squared deviation scores.

The final step, Step 6, requires taking square root of the variance. Standard Deviation = $\sqrt{\text{variance}}$

Standard deviation is the square root of the variance

In the example, $\sqrt{1} = 1$

Why do you take square root of the variance? You do it because you had squared the deviation scores at Step 4. Remember, you had done it because you would have otherwise ended up with zero as variability. You had then squared the deviation scores to get over this difficulty. Now, you would like to remove the influence of squaring the deviation scores in order to obtain the standard deviation. You do this by taking the square root of variance at Step 6.

The concept of standard deviation (or variance) is the same for a sample as for a population, however, the details of calculations differ slightly, depending on whether you have sample data or a complete population. Reproduced below are formula for calculating the sum of squares, variance and the standard deviation of a population of scores.

Sum of squares (SS) is the sum of the squared deviation scores

$$SS = \sum (X - \mu)^2 \text{ or } SS =$$

$$\text{Population variance} = cr^2 = \frac{SS}{N} = \frac{\sum X^2 - (\sum X)^2}{N}$$

$$\text{Population standard deviation} = cr = \sqrt{\frac{SS}{N}}$$

E.g. For the following population of scores calculate SS, Variance and Standard Deviation

2, 10, 6, 9, 8, 7

13.5.4 Standard Deviation and Variance for Samples

Variance and standard deviation for sample data have the same basic definitions as they do for populations: To compute these, we first will need to find SS, the sum of squared deviations. The formulas we use to compute sample SS are essentially identical to the formula used for populations.

Definitional formula: $SS = \sum (X - \bar{X})^2$

$$\text{Computational formula: } SS = \frac{\sum X^2 - (\sum X)^2}{N}$$

Note that the difference between these formulas and the population formulas is minor. We have substituted $\bar{X}$ in place of μ and n in place of N . The difference in notation will have no effect on the calculations.

In order to correct the bias in sample variability, it is necessary to make an adjustment in the formulas for samples variance and standard deviation. This is done by replacing n with $n-1$:

$$\text{Sample variance} = S^2 = \frac{SS}{n-1}$$

Sample standard Deviation (S) is simply the square root of the variance.

$$\text{Sample standard Deviation} = S = \sqrt{\frac{SS}{n-1}}$$

Dividing the sum of squares in the numerator by $n-1$ instead of n , increases the value you will obtain. The adjustment makes sample variability an accurate, or unbiased, estimator of population variability. For this reason, sample variance is often called estimated population variance and the sample standard deviation is called estimated population standard deviation.

Check Your Progress: 4

Find the Range, Quartile Range, Variance and Standard Deviation of the following:

- 1) 5, 14, 13, 9, 7, 7, 8, 6, 12

.....

.....

.....

2) 15, 18, 9, 23, 18

.....

.....

3) 9, 3, 9, 6, 6, 9

.....

.....

4) 28, 39, 42, 29, 39, 40, 36, 46, 41, 30

.....

.....

5) 133, 215, 250, 108, 206, 159, 206, 178

.....

.....

6) 76, 94, 76, 82, 78, 86, 90

.....

.....

7) 52, 61, 49, 52, 49, 52, 41, 58

.....

.....

8) 58, 59, 63, 45, 44, 41, 36, 36

.....

.....

9) 29, 29, 31, 41, 42, 43, 31, 29, 23, 25

.....

.....

10) 59, 54, 37, 45, 31, 60, 32, 34

.....

.....

13.6 LET US SUM UP

In this unit we explained the basic analytical tools under the descriptive statistics category. We discussed the fundamental concepts of quantitative research. The basic concept of quantitative techniques is to make generalisations of research findings based on a group of samples. Statistics plays a crucial role in the generalisation process.

We also discussed the building blocks of measurements - Nominal, Ordinal, Interval and Ratio. Before collecting any quantitative data for a study, researcher

should decide the nature of statistical applications s/he is going to employ; that determines the nature and need of scale of measurements in data collection instruments. These measurements would generate a different set of data, it was discussed how it can be classified as continuous and discrete variables.

It was further outlined how these data can be classified and summarised in the form of frequency distribution with the nature and types of graphical presentation of data, measures of central tendency - mean, median, and mode; measures of dispersion - range, quartile, variance and standard deviation.

It is hoped that you will be able to use above statistical methods while analysing your data when embarking on your research journey.

13.7 REFERENCES AND FURTHER READINGS

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13.8 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Nominal
- 2) Ordinal
- 3) Ratio

Check Your Progress 2

1.	S.No.	Age Intervals	Male	Male%	Female	Female%
	1	Under 20	0	0	100	16
	2	21-25	17	4	83	13
	3	26-30	40	8	60	10
	4	31-35	35	7	65	10
	5	36-40	51	11	49	8
	6	41-45	48	10	52	8
	7	46-50	55	12	45	7
	8	51-55	56	12	44	7
	9	56-60	53	11	47	8
	10	61-65	56	12	44	7
	11	Over 65	67	14	33	5
	Total		478	100	622	100
	478+622=1100			43.5		56.5

2.	S.No.	Time Intervals	Girls	Girls%	Boys	Boys%
	1	140-149	0	0.00	0	0.00
	2	130-139	0	0.00	32	21.05
	3	120-129	0	0.00	48	31.58
	4	110-119	1	1.96	29	19.08
	5	100-109	0	0.00	18	11.84
	6	90-99	3	5.88	14	9.21
	7	80-89	5	9.80	5	3.29
	8	70-79	6	11.76	5	3.29
	9	60-69	14	27.45	0	0.00
	10	50-59	7	13.73	1	0.66
	11	40-49	11	21.57	0	0.00
	12	30-39	4	7.84	0	0.00
		Total	51	100.00	152	100.00
		51 + 152 = 203		25.12		74.88

Check Your Progress 3

Q.No	Mean	median	mode	Q.No
1	40	38	28	1
2	51.4	45	53	2
3	58.4	58	56, 58	3
4	52.6	54	56	4
5	53.4	56	56	5

Check Your Progress 4

Q.No	Range	Quartile Range 0%, 25%, 50%, 75%, 100%	Variance	Standard Deviation
1	5, 14	5, 7, 8, 12, 14	10.5	3.24
2	9, 23	9, 15, 18, 18, 23	26.3	5.13
3	3, 9	3, 6, 7.5, 9, 9	6	2.45
4	28, 46	28, 31.5, 39, 40.75, 46	37.1	6.1
5	108, 250	108, 152.5, 192, 208.25, 250	2183.84	46.73
6	76, 94	76, 77, 82, 88, 94	50.48	7.10
7	41, 61	41, 49, 52, 53.5, 61	36.5	6.04
8	36, 63	36, 39.75, 44.5, 58.25, 63	115.36	10.74
9	23, 43	23, 29, 30, 38.5, 43	51.12	7.15
10	31, 60	31, 33.50, 41, 55.25, 60	149.14	12.21

UNIT 14 DATA ANALYSIS

Structure

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14.0 INTRODUCTION

Data analysis is defined as the processing of collecting raw data and converting it into information useful for decision-making by users. The data are analysed to answer questions, test hypotheses, prove or disprove theories. The methods cover diverse operations of categorising, segmenting and interpreting evidence or data sets.

The previous unit, i.e. Unit 14 explored and explained basic statistical analysis at length. You should look at this unit alongside Unit 14 because here we try to understand the underlying principles of data analysis.

The process of data analysis by the researcher begins when entire set of the data have been collected. Organising the data correctly can save a lot of time and prevent mistakes. Most researchers choose to use a database or statistical analysis programme so that they can format data to fit their needs in order to organise their data effectively. Each data analysis software has its own requirements for how data are to be entered. A good researcher enters all of the data in the same format and in the same database, as doing otherwise might lead to confusion and difficulty with the statistical analysis later on. Sometimes, there are issues of interoperability between different software. Once the data have been entered, it is crucial that the researcher check the data for accuracy. This can be accomplished by spot-checking a random assortment of participant data groups, but this method is not as effective as re-entering the data a second time and searching for

discrepancies. This method is particularly easy to do when using numerical data because the researcher can simply use the database programme to sum the columns of the spreadsheet and then look for differences in the totals. One of the best method of accuracy checking is to use a specialised computer programme that can cross-check double-entered data for discrepancies (as this method is free from error), though these programmes can be hard to come by and may require extra training to use correctly.

In this unit, we shall focus on the concept of data analysis in research and the process of organising and coding of data for effective analysis. We shall discuss how to distinguish between quantitative and qualitative data analysis and so that you are able to apply appropriate data analysis techniques in your research work.

14.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- describe the importance of data analysis in research;
- organise data for effective analysis;
- explain the importance of data analysis;
- distinguish between quantitative and qualitative data analysis;
- explain how to handle quantitative data; and
- discuss how to handle qualitative data.

14.2 DIFFERENT RESEARCH PERSPECTIVES

In the previous units, you have read that the field of media and communication research is characterised by a variety of different research perspectives. The hybrid nature of the field of empirical enquiry in research presents a special challenge for interfacing quantitative and qualitative research analysis.

The understanding of the term 'analysis' has tended to divide qualitative and quantitative research traditions. In quantitative media studies, rule governed procedures of segmenting and categorising the components of content, form the basis of interpretation. On the other hand, qualitative studies typically emphasise meaning as a process unfolding in contexts.

The use of the words 'quantitative' or 'qualitative' can refer to a broad approach to the research problem, a methodology for the research and a defined set of research techniques. When we refer to quantification the focus of the research analysis lies on numbers and, in the case of content analysis, on measurable numerical values for 'space' and 'time'. These appear more systematic and replicable. Survey research also relies on greater quantification and greater measurement sophistication. However, this sophistication comes with a price: increasing quantification narrows the types of questions that can be addressed. This means that research depth is sacrificed to get research breadth. Basically, in quantitative data, we get a lot of findings that can be generalised to a larger population; while in qualitative data analysis, we get depth and underlying causes, without being able to generalise to a larger population.

For this reason, there is no commonly accepted definition of the term ‘qualitative’. Qualitative research relies mainly on observations and verbal data that reflect everyday experiences. On the other hand, and consequently, qualitative data have to be analysed differently.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

A. Fill in the blanks:

- 1) is defined as the processing of collecting raw data and converting it into information useful for decision-making by users.
- 2) The data is analysed to answer questions, test. or disprove theories.
- 3) Once the data has been entered, it is crucial that the researcher check the data for.....
- 4) In quantitative media studies, rule governed procedures of and the components of content, form the basis of interpretation.
- 5) Qualitative research relies mainly on and verbal data that reflect everyday experiences.

14.3 HANDLING QUANTITATIVE DATA

Quantitative research methodologies generate numerical data. Surveys (whether of audiences or content) and experiments are the basic ‘methods’ of the data collection. Once numerical data have been collected, they need to be analysed through statistical techniques. These mathematical techniques are used to describe, organise as well as explore relationships within the data. In epistemological terms, quantitative research is typically grounded in a hypothetico-deductive approach, in which investigators mount hypotheses or predictions about the expected associations or cause-effect relationships between variables. The aim of the quantitative data collection and analysis, then, is to produce findings which lead to the acceptance or rejection of a specified hypothesis. Numerical data analysis through statistical procedures represents a systematic and objective way of determining whether significant patterns of relationships exist among those phenomena that have been measured in data collection.

14.3.1 Organising and Coding Data

Numerical data is collected through survey questionnaires, content coding frames or experimental instruments. These have to be systematically coded and entered into a database for further analysis. This is the process of organising and coding data and is a very crucial part of the analysis process. This is where most inexperienced researchers tend to make mistakes. If there are errors here, they will be reflected in the problems faced while analysing data and will make analysis difficult, if not impossible.

The first stage of organising data is the creation of a coding manual. Often, researchers think that data can be entered directly into a database. While, yes it can be entered, the coding manual anticipates any problems in entering, e.g. when there are multiple answers possible, or when there are double digit replies.

Simply stated, a coding manual is a set of instructions on how the data are to be tabulated, and is an extremely important document when organising data, especially, if there is more than one person entering data onto a coding sheet or database.

You may ask, what does a coding manual look like. A brief design is given below:

Variable	Numerical value	Column
Location	1) Rural 2) Semi-urban 3) Urban 9) No Answer	1
Sex	1) Male 2) Female 3) Transgender 9) No Answer	2
Level of Education	1) Illiterate 2) Upto 8th standard 3) Matriculate 4) High School 5) Graduate and Above 9) No answer	3

Note, how one column is assigned to one variable. And since a single digit cannot exceed 9, 9 is the code given for a 'no answer'. In case there are more than 9 possible answers to a question, assign two columns, with 99 for a 'no answer'.

Sometimes, a question may have multiple answers because that is how it has been asked. Then, one column must be assigned for each possible answer.

A sample coding sheet is provided in the Annexure. In a coding sheet, one row is assigned to one respondent. One or two columns (as the need may be) are assigned to a question or variable. Through this process of coding, identification of key variables can be done easily and errors, duplication, and repetition etc. can be addressed. Once the data are coded numerically and entered into a coding sheet and transferred onto a computerised database, data can be checked for accuracy and statistical tests can be performed. The accuracy with which the data transfer takes place and the application of statistical procedures that are appropriate for the particular type of data, are crucial to the entire quantitative research work—errors in the early stages as it can skew or invalidate the final results.

Different software packages have different methods for entering data into columns. For instance, in Excel, it is possible to enter two numbers in one cell separating them as 1, 3. In SPSS, one cell in a column can have only one number. Follow the instructions of the software package in designing your coding manual and in entering your data

14.3.2 Descriptive Statistics

Descriptive statistics has been discussed in detail in Unit 14, we will merely summarise different types of statistics here. Often quantitative data analysis begins by adopting a simple descriptive approach in order to establish some initial patterns in the findings. For instance, a survey of the public opinion about the performance of elected political leaders might first present data in percentages of the number of respondents who agreed or disagreed that the political leaders were competent. In a further analysis, results can be described in terms of gender, age, social class and political affiliations of the respondents. Such results can be visually displayed in a bar chart or summarised in a table. Some of the ways of displaying results are described.

Another type of study might conduct a survey to ask a sample of 1000 respondents to state how many hours are spent on watching television each week. In this case, descriptive statistics can be applied to show how many respondents viewed nothing or less than an hour or between one to two hours and so on. Next, a frequency distribution could be generated showing how the respondents were distributed across different volumes of viewing. Such data can be visually represented in a line graph or bar chart, or a simple table.

14.3.3 Summary Statistics

Data can be further analysed in terms of summary statistics. Summary statistics measure two basic aspects of the distribution of 'scores' or measurements in a data set: central tendency and dispersion, or variability. A central tendency measure indicates which out of a range of scores is the 'typical' one. This typical score in turn can be defined in three different ways:

- a) The mode is the most frequently occurring in a range of scores. For example, if in a set of ten scores, five score '4', three score '2', and two score '1', the mode is 4.
- b) The median score is the midpoint in a range of scores. In the following set of scores the median is "7": 2 4 5 6 (7) 9 10 11 12. The score '7' lies at the exact half-way point in this distribution of scores. In other cases where there is an even number of scores and therefore no exact mid-point, the median must be calculated by averaging between the two centre scores: 4 5 7 8 9 11 12 13. Here, the median is 8.5 or the average of '8' and '9'.
- c) The mean score is the average of the total range of scores. In the previous example the eight scores totalled 62, which divided by 8, gives a mean score of 7.75.

14.3.4 Descriptive Measures

Other fundamental descriptive measures are range, variance and standard deviation.

Range, which is the simplest expression of dispersion, is the difference between the highest and the lowest scores in a particular distribution.

Variance provides a mathematical index of the degree to which scores deviate from the mean score, and tend to be expressed not in terms of the original scores, but as squared deviations from the mean. To compute the variance, one subtracts

the mean of a distribution from each score and then squares the result. These squared scores are then summed and divided by the number of original scores minus one. Variance is a powerful and widely applied measure.

Standard Deviation is a statistical procedure that enables us to see by how much the members of a group differ from the mean value for the group. A low standard deviation means that most of the members of the group are very close to the average, while a high standard deviation score means that the numbers are spread out. Unit 14 has explained the procedure of arriving at a standard deviation at length.

14.4 TESTING HYPOTHESES

Much quantitative research goes beyond the simple description of data and their distributions. The search for connections or correlations to understand the underlying causes of behaviour is an important part of any data analysis. For this, the researcher may have framed specific questions or hypotheses. The data have to be tested to either validate or disprove the hypotheses framed at the beginning of the research. In hypothesis testing, the researcher is interested in establishing whether two or more variables are associated, or whether the scores obtained in two or more groups are different - and if so, whether such association is in a 'significant' manner. Significance means simply that the association of the variables must not happen by chance, or the laws of probability alone. When a relationship is highly significant, it means that such an association would not have happened in the normal course of things, and that there is an important association between the variables.

When deciding whether to accept or reject a hypothesis, the researcher must examine the statistical significance of the results. The starting point is to set up a null hypothesis, or a hypothesis which asserts that any statistical differences or relationships that emerge within the data are entirely due to chance fluctuations or random error. The research hypothesis puts up an alternative viewpoint- that the statistical relationships or differences are not due to chance, but represent real phenomena that can be explained theoretically in the research.

The essential characteristic of quantitative research is that it reduces phenomena to numerical codes. Numerical measurement, however, can occur at more than one level, and the understanding of different levels of measurement is crucial, both to the correct use of statistical methods of data analysis and to the proper interpretation of data.

Within mass communication research, hypothesis testing is done using tests of association and tests of significance. The term "association" is closely related to the term 'correlation'. A commonly used test of significance is the chi-square test for independence. Also called the Pearson's chi-square test or the chi-square test of association, this test is used to discover if there is a relationship between two categorical variables. Very simply, the test may show us if there is a relationship between the sex of a respondent and the amount of television s/he watches, or his/her social media habits.

Tests of significance are also used in mass communication to tell us how strong a relationship between two variables is. The types of tests of significance commonly used are-T Test, the f-test and analysis of variance (ANOVA).

Essentially these tests tell us about the significance of difference between the mean of a small sample and hypothetical mean of population (expressed in terms of standard error).

At this stage, it is not necessary for you to know the formulae to perform these tests and arrive at some scores. You can programme the software you are using which in sum will calculate the scores for you, give you the score, the P value, and the significance level. What is important for you is to know how to read the scores and interpret them.

The first step is to determine if the association is significant. The second step is to determine which variables have the most impact. To do this, compare the p-value to the significance level on a chart available in any book on basic statistics. For a chi square test, if the significance level is less than .05, you can conclude that an association between the variables exists. Different tests have different significance levels. You should either look up a book on basic statistics, or consult a statistician to get an accurate picture.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

A. Fill in the blanks:

- 1) In epistemological terms, quantitative research is typically grounded in a approach.
- 2) The aim of the quantitative data collection and analysis, then, is to produce findings which lead to the..... or..... of a specified hypothesis.
- 3) The.....is the most frequently occurring in a range of scores.
- 4) The median score is the.....in a range of scores.
- 5)is the difference between the highest and the lowest scores in a particular distribution.

14.5 QUALITATIVE DATA ANALYSIS

Qualitative studies are different as they draw their strength not from numbers but from answering the questions 'why' and 'how'. A quantitative study will provide vast amounts of data for analysis, while a qualitative study will enable in depth analysis.

Qualitative data comes in a variety of forms such as field notes made while observing in the field, interview transcripts, documents, diaries and journals. Also, during the course of the study the researcher accumulates a large volume of data. To organise, analyse and make sense of this data poses special challenges to the researcher.

Unlike the quantitative approach, where the data analysis does not begin until all the numbers are collected, data analysis in qualitative research is a continuous process and is done as early as from the time of collection of data and throughout the project.

Another important dimension of difference between the qualitative and the quantitative method is that while in quantitative analysis hypotheses are developed prior to the study, in qualitative analysis researchers use an inductive method. The data collected in this method are grouped into appropriate and meaningful categories; explanations emerge from the data.

Organising the data

The researcher is the main instrument in qualitative data collection and analysis and therefore must prepare before beginning the task of investigation. Some social scientists describe this preparation as *epoche*, the process by which the researcher tries to become aware of and remove, as far as possible, prejudices, viewpoints, or assumptions that might interfere with the analysis. *Epoche* helps the researcher put aside personal viewpoints so that the phenomenon under study may be seen for itself.

To facilitate working with large amounts of data generated by qualitative analysis, the researcher generally first organises the information along a temporal dimension. This means that the data are arranged in a chronological order according to the sequence of events that have occurred during the process of data collection. Further, each piece of information is coded identifying the source and multiple copies of notes and all other documents are gathered.

Working with vast amounts of data can be very difficult and time consuming. And today, there are a number of software applications which help you to collate and organise the data collected. However, the process, whether done by hand or through the use of software remains more or less the same, and it needs to be understood.

Let us take this process further step by step.

Identify the main themes: Carefully go through all the descriptive responses given by respondents to each question and understand the meaning they communicate. Once these are understood, broad categories or themes can be developed so that data can be classified and responses analysed. From these responses the researcher develops the broad themes that reflect these meanings. These themes become the basis for analysing the text of unstructured interviews.

Assign some codes to the main themes: If the researcher wants to count the number of times a theme has occurred in an interview, s/he needs to select a few responses to an open-ended question and identify the main themes. S/he continues to identify these themes from the same question till a saturation point is reached. Write these themes and assign a code to each of them, using numbers or keywords.

Classify responses under the main themes: Go through all the responses and fit them into the main themes identified.

Integrate themes and responses into the findings: Link the qualitative responses with quantitative measures - either validating the number or developing an understanding of why people responded in a particular way. While discussing the main themes that emerged from their study, some researchers use quotations to keep the feel of the response. Such quotations, especially from case studies add depth to the understanding of the problem. The themes must connect to the objectives and research questions in the report.

It is important to note that, in qualitative research, hypotheses are not tested. It entirely depends on the way the qualitative data are analysed and reported

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

A. Fill in the blanks:

- 1) The three aspects of qualitative empirical research are, and
- 2) In quantitative analysis, hypotheses are developed.....to the study.
- 3) In qualitative research data analysis is a..... process.
- 4)is the process by which the researcher tries to remove or at least become aware of prejudices.
- 5) To facilitate working with large amounts of data, the researcher generally first organises the information along..... a dimension.

14.6 DRAWING CONCLUSION THROUGH DATA ANALYSIS

Qualitative data can be analysed through different techniques. The two popular techniques are: the constant comparative technique and the analytical induction technique.

14.6.1 Constant Comparative Technique

This process involves the following four steps:

- Comparatively assigning incidents to categories
- Elaborating and refining categories
- Searching for relationships and themes among categories
- Simplifying and integrating data into a coherent theoretical structure.

After the data have been prepared for analysis, the researcher places each unit of analysis into a set of provisional categories. As each new unit is examined, it is compared to the other units previously assigned to that category to see whether its placement is appropriate. If some units of analysis do not fit into any pre-existing category, new classifications may be created. Units that fit into more than one category should be noted and placed into a new category by finding similarities among units that fit into a category.

The researcher may write rules or propositions that attempt to describe the underlying meaning that defines a category and they may also be revised throughout the study. This helps the researcher to focus and crystallise the study and explore the theoretical dimensions of the emerging category system.

In the next stage, the researcher may examine the propositional statements and look for meaningful connections. Finally, the researcher writes or summarises the research findings and integrates them into a coherent explanation of the phenomenon in relation to the objectives of the study.

14.6.2 Analytical Induction Strategy

This method blends hypothesis construction and data analysis. In this strategy, the topic of research is defined and the hypotheses are developed. The hypotheses may be refined or reformulated until they can be adequately tested. The researcher uses the analytical induction approach in explaining a phenomenon.

Let us try to understand this with the help of an example. Suppose a researcher is interested in explaining why people watch home-shopping channels. The answer to this may be obvious that people do so in order to buy products. The researcher may not be satisfied but decides to use this as an initial hypothesis. S/he further looks out for a heavy viewer of these channels. During the course of the interview, the respondent may say that the primary reason for watching the channels is to find new and unique products. With this input, the researcher can reformulate the hypothesis. Another viewer may answer this differently by emphasising that s/he uses the prices advertised on the channel to compare before shopping. The hypothesis is further redefined. As the researcher goes further ahead s/he finds cases that do not add to the already existing reasons and hence cannot fit into the revised hypothesis. This method can be exhausting and it can be difficult for the researcher to determine the exact stopping point.

Then what should be done? This is a tricky question for which there is no simple answer. One way, especially when researching an event such as a riot, is to limit one's findings to the main objectives of the research, and to clearly state that this is being done. Another way is to define the limitations or scope of the research early on in the designing phase itself so that it is easy to see what can be included and what cannot. And finally, if there is a time limit on the study (especially if funded by a donor) or as time to submit the draft of the thesis, the time limit will itself indicate when to stop. Do remember, that all research is finite in time - it does not go on endlessly and the researcher has to determine when it is best to stop.

14.6.3 Generalising about Generalisation

The literature and the theory of science recognise two different conceptions of 'generalisation'. Empirical or statistical generalisation refers to the capacity of quantitative methodologies to apply to predefined (hypothetically deduced) categories to a representative set of empirical instances, thus supporting external validity. Theoretical or analytical generalisation refers to the articulation of new concepts or categories, typically in qualitative inquiry, that conceive empirical instances in a more consistent and insightful manner, giving priority to internal validity. Both these aspects are relevant for assessment of findings and insights of both qualitative and quantitative studies.

14.6.4 Converging Practices and Approaches

In recent decades a number of research scholars have outlined ways of combining and mixing quantitative and qualitative methodologies. Three principal forms have emerged as possible methods of combining qualitative and quantitative methodologies. These are facilitation, triangulation and complementarity.

Facilitation: The most common practice traditionally has been to treat qualitative and quantitative approaches as relatively separate steps in research. Facilitation

aims at facilitating in-depth analyses of respondents by formulating interpretive categories. For example, in survey research, analytical strategies are arrived at and that are conceptually precise and meaningful to the respondents. The mapping of a social life of respondents may be facilitated by subsequently drawing up interpretative categories after analysing the members in-depth.

Triangulation: It is a strategy for gaining several perspectives on the same phenomenon. It can be carried out through several data types, several investigators, and several methodologies. For example, experimental and observational approaches can be combined and analysed for interpretation of a phenomenon.

Complementarity: Here, different analytical categories and procedures may be appropriate for capturing particular aspects of the same empirical domain, or for addressing two different domains with a bearing on the same research question. In the final stage, two categorically different sets of findings can be joined with reference to a common theoretical framework.

Check Your Progress: 4

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

A. Fill in the blanks:

- 1) The two popular techniques of qualitative data analysis are the..... and technique.
- 2) In the constant comparative technique, the researcher places each unit of analysis into a set of categories.
- 3) The method of analytical induction blends and data analysis.
- 4) Three principal forms of combining qualitative and quantitative methodologies are , and
- 5) is a strategy for gaining several perspectives on the same phenomenon.

14.7 LET US SUM UP

In this unit we have discussed the purpose, importance and significance of data analysis in research. Data analysis is defined as the processing of collecting raw data and converting it into information useful for decision-making by users. The data is analysed to answer questions, test hypotheses or disprove theories. Various methods that are applied in data analysis - the quantitative and the qualitative have been discussed in this chapter.

The understanding of 'analysis' has tended to divide qualitative and quantitative research traditions. In quantitative media studies, rule governed procedures of segmenting and categorising the components of content, form the basis of interpretation. While, qualitative studies typically emphasise meaning as a process unfolding in contexts. Ultimately, it is not an either/or situation. Each tradition has to feed into and supplement each other in order to yield accurate results and logical explanations for phenomena. The way data are gathered and analysed may differ, but in the long run, both aim to understand a situation or solve a problem.

The aim of the quantitative data collection and analysis is to produce findings which lead to the acceptance or rejection of a specified hypothesis. Numerical data analysis through statistical procedures represents a systematic and objective way of determining whether significant patterns of relationships exist among those phenomena that have been measured in data collection. Descriptive and summary statistics and descriptive measures are used to analyse the data.

Qualitative data comes in a variety of forms such as notes made while observing in the field, interview transcripts, documents, diaries and journals. The constant comparative technique and analytical deduction strategy are applied to analyse and draw conclusions from the data.

In recent decades, a growing number of scholars and researchers have suggested and outlined ways of mixing and combining the quantitative and the qualitative methodologies. These combining forms can enrich and reflect in-depth and breadth the status of that knowledge.

With this unit, you have almost come to the end of the research process. The final stage is report writing or the presentation of your research for your peers, the academic community, the funding agency, or simply for your next academic degree. The last unit of this course explains how to write your report effectively.

14.8 KEYWORDS

Arthur Asa Berger, 2011, *Media and Communication Research Methods*, Sage Publications.

Anders Hansen & David Machin, 2013, *Media and Communication Research Methods*, Palgrave Macmillan.

Klaus Bruhn Jensen, 2012, *A Handbook of Media and Communication Research: Qualitative and Quantitative Methodologies*, Routledge, London.

Roger D. Wimmer and Joseph R. Dominic, 2010, *Mass Media Research, An Introduction*, Wadsworth.

14.9 REFERENCES AND FURTHER READINGS

Data analysis: It is the process of collecting raw data and converting it into information useful for decision-making by users.

External validity: The degree to which the results of a study are generalisable to other situations.

Epoche: It is the process by which the researcher tries to remove or at least become aware of prejudices.

Hypothesis: A tentative generalisation about the relationship between two or more variables that predict an outcome.

Internal validity: The properties of a research study are such that the results are based on expected conditions rather than on extraneous variables.

Operationalisation: A definition that specifies patterns of behaviour and procedures to experience or measure a concept.

Summary statistics: Statistics that summarise a great deal of numerical information about a distribution, such as mean, standard deviation.

Skewness: The degree of departure of a curve from the normal distribution (curves can be positively or negatively skewed).

Triangulation: It is a method using a combined quantitative and qualitative approach to solve a problem.

Qualitative research method: A research method that uses flexible questioning.

Quantitative research method: A research method that uses standard questioning.

14.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- 1) Data Analysis
- 2) Hypotheses
- 3) Accuracy
- 4) Categorising, Segmenting
- 5) Observations

Check Your Progress 2

- 1) Hypothetico-deductive
- 2) Acceptance, rejection
- 3) Mode
- 4) Mid-point
- 5) Range

Check Your Progress 3

- 1) Strategy, Tactics, Techniques
- 2) Prior
- 3) Continuous
- 4) Epoche
- 5) Temporal

Check Your Progress 4

- 1) Constant comparative, the analytical induction
- 2) Provisional
- 3) Hypothesis construction
- 4) Facilitation, triangulation and complementarity
- 5) Triangulation

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UNIT 15 REPORT WRITING

Structure

- 15.0 Introduction
- 15.1 Learning Outcomes
- 15.2 Stages in Report Writing
- 15.3 The Beginning
 - 15.3.1 The Title
 - 15.3.2 Abstract
 - 15.3.3 Acknowledgements
 - 15.3.4 List of contents
 - 15.3.5 Introduction
- 15.4 Main Body of the Report
 - 15.4.1 Literature Review
 - 15.4.2 Objectives
 - 15.4.3 Methodology
 - 15.4.4 Sampling and Sample
 - 15.4.5 Research Instruments
 - 15.4.6 Data Presentation
 - 15.4.7 Findings and Analysis
- 15.5 The Final Section
 - 15.5.1 Summary
 - 15.5.2 Conclusions
 - 15.5.3 Recommendations
 - 15.5.4 Appendix
 - 15.5.5 References
- 15.6 Effective Writing
- 15.7 Let Us Sum Up
- 15.8 Keywords
- 15.9 References and Further Readings
- 15.10 Check Your Progress: Possible Answers

15.0 INTRODUCTION

Report writing is both an art and a craft. In the previous Blocks and Units you have learnt the whole process of communication research, research methods, and data analysis. This unit focuses on bringing it all together in the form of a report that reflects the work you have done and to present the report to the reader, detailing what the report means and what you have found during the research process.

A report is a structured, formal document written to convey factual, research based information. This information is often used as basis for making decisions and taking action. Research reports generally follow a particular standardised structure and format which are described and discussed in this unit.

15.1 LEARNING OUTCOMES

After reading this unit, you should be able to:

- outline major components of a research report:
 - discuss various stages in research report writing; and
 - write a good research report.
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15.2 STAGES IN REPORT WRITING

The following stages are involved in writing a report:

- Planning the report
- Organising the information
- Structuring the information
- Writing the first draft
- Checking and revising
- Writing the final version

Planning your work: You will be writing a report for a reason. The best reports are the ones that are clear and purposeful; so you need to ask yourself:

- Who am I writing for?
- Why am I writing this?
- What am I trying to say?
- How should I be saying it?

A report usually falls into one of the following categories:

Informative - presents statements of fact and opinions.

Persuasive- tries to obtain approval from the reader or advocates a position.

Explanatory - answers given questions and gives a conclusion. Operational - records an event and gives evidence of work done. Which category does your report fall into?

Activity-1

Answer the four questions for yourself: Who am I writing this report for? Why am I writing this? What am I trying to say? How should I be saying it?

A report informs the readers, and, if it is logically presented, can be extremely useful to both the reader and the writer. It is a more structured form of writing than say, an essay, and is designed so that it can be read quickly and accurately; though reports may not be necessarily read in a consecutive order. Structure and convention in written reports stress the processes by which the information was gathered as much as about the information itself. Reports should be organised for the convenience of the intended reader. These are written on a wide range of subjects for a wide variety of reasons. A research report follows the general principles and standard structure of the general report writing.

Format of the Report

Over the years, a standard format for reports has been worked out. Although there may be circumstances when it is advisable to change the format to fit a particular need, generally, the structure is common. The standard sections of a report are as follows:

The Beginning

Main body of the report

The Final section

Let us discuss each section in detail.

15.3 THE BEGINNING

The Beginning of the report contains the following: Title Page, Abstract or Executive Summary, Table of contents, List of tables, List of figures and Introduction. Let us take up each section of a report in greater detail.

15.3.1 The Title

The report title is the first thing a reader sees. It tells them of the report's content, and should thus be precise, clear and concise; and should also give the reader ample information to differentiate it from other reports on similar subjects.

15.3.2 Abstract

An abstract, also sometimes called the Executive Summary in a report to an organisation, is an independent text, written in a brief, non-repetitive style that contains the essential details of the research presented in the report. It offers a quick way for the reader to decide whether the report is relevant for him/her and whether it contributes important knowledge/key results/valuable scientific methods. An abstract is usually written for a wider readership than the report. Typically you should not include extensive background, literature review, detailed methods, figures, or references. Also, generally you should not add any material which is not in the report. For short reports, about 150 words are enough; for a longer report, the abstract should not exceed one page. An executive summary may be longer, but it is never more than two or three pages. Typically, abstracts of research reports, including dissertations, are often made publicly available on the Internet, enabling anyone with an interest in the subject to review and see if the content is relevant to their research.

15.3.3 Acknowledgements

Acknowledgements are an important part of the research report where you can thank those who have helped you in any way during the research work. In an academic report, they are generally your research guide, the faculty, administrative and support staff of the department. In addition to persons you may thank organisations, institutions or the libraries you had visited for material collection. They can also be people who have extended you emotional support such as your family members, friends, near and dear ones. You should not forget to thank the respondents if your study is based on field research as without them, there will be no study. You should be generous in your approach but do not gush and make

it very sentimental. Using a variety of words and expressions for different set of people adds to the impact of your acknowledgment section.

15.3.4 List of Contents

The list or table of contents helps the reader find specific sections of the report at one glance. The contents are presented at the beginning of the report as a list of section and sub-section headings along with the corresponding page numbers.

15.3.5 Introduction

This section introduces the research in a broad way by outlining its rationale, aims and objectives. The purpose of the introduction is to set the scene for your readers so that they know what to expect from the report. The introduction answers the following questions:

- Why is the report being written?
- What kinds of information does it contain?
- How is the problem being approached?
- For whom is the report being written?

It is often useful to try a three-paragraph approach to your initial report writing:

Paragraph 1 - What it is you are investigating, why it is interesting?

Paragraph 2 - What information is already known on the topic?

Paragraph 3 - What were the aims of the study you have just done?

Make sure you connect the paragraphs together with a linking sentence following the fundamentals of good writing.

Check Your Progress: 1

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) What are the three key sections of a research report?

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2) Describe the sub sections in the three key sections of the research report?

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15.4 MAIN BODY OF THE REPORT

The body of the report includes several important sections - theoretical framework, literature review, research design, objectives, methodology, sampling and sample, research tools, data collection process, findings and analysis, and interpretation.

The body presents the actual content of the report and describes the subject matter. Being the most extensive section of a report, it answers several questions like, “What did I do?”, “How did I do it?”, and “What did I do it with?”, and “What did I find?”. The section should have detailed information on what you have done so that the reader understands it clearly, and can replicate the tests if required, and check their validity.

Let us now understand some of the important sections of the main body of the report in detail

15.4.1 Literature Review

In this section of the report, one locates and reviews the articles, reports, documents which are near or about the research topic or question under consideration. This section also gives an insight into the theoretical framework that may have been used previously and how that could help you relate, generate and base your research on a sound theoretical base. In Block 1, Unit 5, we discussed how the review of literature is done. In this section of your report, you must present the findings of the survey of literature, summarising previous research, and place the review in the context of your own work. From this context, you can connect your research to previous work, clarify the gaps in the field, and justify your own research. This section is not a listing of what research was done by author or title; rather it is a cogent argument connecting your work to that of others. The survey of literature is organised around ideas and themes, and should conclude what you have learned from reviewing the literature so far. After reading your review your reader should be convinced that your proposed research project will play a necessary role in furthering knowledge in your field

15.4.2 Objectives

The section on objectives draws the attention of the reader to the specific issue under study. Though each research study has its own specific purpose, we may think of research objectives as falling into a number of following broad groupings:

- To gain familiarity with a phenomenon or to achieve new insights into it. Studies with this object in view are termed as exploratory or formative research studies.
- To portray accurately the characteristics of a particular individual, situation or a group. Studies with this object in view are known as descriptive research studies.
- To determine the frequency with which something occurs or with which it is associated with something else. Such studies are known as diagnostic research studies
- To test a hypothesis of a causal relationship between variables. These type of research studies are called hypothesis-testing research studies.

The objectives of the given research must be stated in clear and unambiguous terms. Following the statement of objectives, the hypotheses to be studied should also be stated clearly.

15.4.3 Methodology

The first section of the Methodology chapter provides a broad picture of the approach to the study. Here you need to state whether the study is exploratory, historical, descriptive, ethnographic, or experimental. This is because each approach carries with it a set of methods that are usually followed. Once you have defined the approach, you can describe what kind of methods and tools you will be using.

15.4.4 Sampling and Sample

Here you need to describe the population of the study as explained in Units 4 and 14. Then, describe the sample size, what kind of sampling procedure was followed, and how exactly the sample was drawn and why. A flow chart, at this stage, would give a graphic description of the sampling process. You should also outline what would be the sampling error or confidence interval (discussed in Block 1, Unit 4). Describe what problems were identified, how they were addressed in sampling from the population to help the reader understand the context of the sampling. For example, in an experimental design, you will have to describe how you selected the subjects and how you placed them in either the control or experimental group.

15.4.5 Research Instruments

Before describing your research instruments, the report must state what the 'unit of study' is and what is the 'unit of analysis'. Let us understand this with an example. If you are conducting a survey, then the unit of study is the individual respondent, whether it is a questionnaire or an interview schedule. However, when you report your data, you will be reporting results in terms of broad categories and groups. These broad categories are your unit of analysis.

The next step is to describe your research instruments, i.e. the questionnaire, the interview schedule, attitude scale, or the knowledge test (in an experimental design). Describe the structure of the instrument and what each section contains. Place the research instruments in the Appendix.

You also need to describe how the instruments were pretested and administered by mail, telephone, online, as individual interviews, or in groups of people and/or when the knowledge test or attitude scale was administered. You may mention any limitations you faced when you actually went through the process of data collection so as to place the research you have done within a context.

Mention any limitations you faced in the field during data collection. For instance, these could be smaller samples, low response rates on the survey, resistance from respondents to answer your questions, or even problems of human resource and logistics which may have impacted the overall findings. Writing this here helps the reader to understand why the research may have deviated from the original design, and it helps to place the findings in context.

15.4.6 Data Presentation

Describe the data processing operations in terms of sifting, checking for accuracy, classification and categorisation.

- **Sifting-** a process of examining the collected raw data to detect errors and omissions and to correct these when possible.
- **Classification-** the process you followed in arranging data in groups or classes on the basis of common characteristics.
- **Categorisation according to attributes-** here data is analysed on the basis of common characteristics which can either be: descriptive such as literacy, sex, religion etc. or numerical such as weight, height, income etc. You can also include the levels of data used (discussed in Unit 14), whether the data were nominal, ordinal, interval or ratio.

Such classification can be either:

- **Simple classification:** where we consider only one attribute, and divide the universe into two classes - one class consisting of items possessing the given attribute and the other class consisting of items which do not possess the given attribute.
- **Manifold classification:** Here we consider two or more attributes simultaneously, and divide the data into a number of classes.
- **Classification according to class-intervals:** is done with data relating to income, age, weight, tariff, production, occupancy etc. Such quantitative data are known as the statistics of variables and are classified on the basis of class- intervals.

You also need to describe the kind of data analysis methods used, and the statistical procedures followed in the study. In this section, you would describe what statistics you used, for example, are they just frequency distributions describing the data or did you test the data for association or significance. If you tested the data, you should describe what tests you used, and how you calculated significance. In this section, you need to justify your choice of statistics and/or in the case of qualitative data, your choice of the way in which you analysed the data and are presenting your data.

15.4.7 Findings and Analysis

In this part of the report you need to present all the evidence that has been collected as a part of your research. You should ensure that the information is organised logically so that conclusions can be drawn naturally from the facts presented. The findings may be structured in the following ways.

In the first part of findings and analysis section, describe the real sample. The final sample of your study is never the original sample drawn; the real sample is always lower than what you planned. If you have a 15 per cent response rate on your questionnaire based survey, say so. The quality of your findings depends on this information.

- You might want to present the demographics in one or many tables. If possible, collapse the demographics table into one where a full picture of

the sample emerges. Always present a table first and then describe and explain the findings in terms of the most important factor first and proceed to the lesser. For instance, “70 per cent of the sample agreed that. . .” and then, proceed in terms of lesser percentages and numbers. Give both percentages and frequencies (n) here.

- The second section of the findings and analysis section focuses on the research questions and hypotheses being tested. Again, present your data in a sequence of each hypothesis listed, the tables showing the results (measures of central tendency, dispersion, variation, association and significance.) Display the table first, and then analyse the results in terms of whether the hypothesis is upheld or disproved. Never place a table without an explanation.
- Tables, graphs and diagrams are useful to present information in a manner that is easily understood. If the information being presented through these tools is necessary to prove the main objective of the study, then it is prudent to place it within the main text or the body. In case it is not, then these can be placed at the end of the report in an appendix. Also, you should ensure that tables and diagrams are properly labelled and the information is presented in a systematic manner, so as not to confuse the reader.
- Do not report unnecessary data. Just because you have collected vast amounts of information during your field work and analysed it, it does not mean you have to report all of it. Excess information will clutter your findings. You can use the additional data for some other research paper.
- Do not include any opinions in this section - report only what you find. Your opinions do not belong here; they will only distract the reader from the main objectives and hypotheses of your research study.

As the writer, it is your job to present the facts and evidence related to the subject of your study to the readers. With that in view, it is not enough to just present a graph or table and expect your reader to pick out what is important. You must, therefore, include some discussion of the facts, but to repeat, no opinions or judgments about the findings.

In this section, discuss your findings in terms of their relationship to the objectives of your study, previous literature on the subject. You can include what may have been the reasons for similarities or differences between previous research studies. It is important that you link your study to the body of theory and research already conducted because it brings continuity and the cumulative nature of research adds value to the hard work you have done.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List the sub-sections of the Main body of the report.

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2) What is a literature review?

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3) What are the different ways of data presentation?

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15.5 THE FINAL SECTION

The final section of the report presents a summary and conclusions of the report. It also comprises recommendations, appendix, references and/or bibliography.

15.5.1 Summary

In this section, you should summarise your report in terms of what you had set out to do, what you did, how you went about your research, and what were your findings - be brief here, no need to elaborate. Readers often look at this section first before proceeding to the main body of your work, so all necessary information should be repeated here.

15.5.2 Conclusions

The conclusions should arise naturally from the evidence that is presented in the previous sections. You may include, for example, statements on what has happened, what the situation is and what might happen. You may also consider further action that could be taken, and an analysis of the advantages and disadvantages of various courses of action. If appropriate, you should give your recommendation, in the light of your experience and the evidence that you have presented, of what a preferred course of action would be. If there are any gaps in your knowledge which prevent you coming to a conclusion, then you should make that clear and perhaps outline what further investigations would be necessary to provide fuller information. You should not include any new information that does not appear in the main body of the report and do not make statements that you cannot support with the evidence

15.5.3 Recommendations

It is not always necessary to present recommendations in a report, but if they are included, they should flow logically from the findings of the study and the conclusions. Usually the recommendations are set out as brief statements and suggestions for further research.

15.5.4 Appendix

The appendix is the section of the report that contains all the secondary or marginal information which does not directly influence the central objective of your study. This information is presented separately so as not to overload your main argument. Texts of questionnaires, interview questions and transcripts, specification sheets, etc. can be presented in the appendix if needed. Some of the other materials that may be included are as follows:

Glossary: It contains the meanings and explanations of the technical terms that have been used throughout the study and is best presented in an Appendix. A glossary of terms is especially useful when a technical study is being presented to people who may not be subject experts. The terms and their concise definitions are listed alphabetically in this section. You may also include acronyms, i.e. short forms for names of organisations in this section.

Tables: While presenting a study with a lot of statistical data, it is not always possible to present all of it in the main body. In such a scenario the data is placed together in an Appendix. While presenting data through tables in Appendix, it is important to give them clear numbers and titles and present them in the order of their reference in the text.

Diagrams and Illustrations: Like tables, diagrams can also be presented within the main text. But, if the writer feels that this disrupts the text too much, the diagrams and illustrations may be presented in the Appendix.

15.5.5 References

All references should be given in full and in alphabetical order. For detailed guidelines on punctuation, capitalisation, and abbreviation in reference citations, see APA, MLA, Harvard, Chicago, CSE or other such relevant manuals. The organisation for whom the report is being prepared will have their own style manual. Ask for it and follow it scrupulously.

Book: the author's surname is written first, followed by the initials or the full first name; date of publication; complete title of the book; the edition; place of publication; and the publisher.

Journal article: the author's surname with the initials or the complete first name; date of the journal's publication; full title of the article (in quotation marks); journal name (in italics) volume and issue number of the journal; the page numbers on which the article can be found.

Electronic or Website reference: The general principle in APA style is to include the same information as you would for a print source, in the same order, and add just enough information about finding the source electronically to help the reader of your paper find that reference. If an online source has a digital object identifier (DOI), a permanent identifier for an electronic publication, you should use that instead of a URL. Also cite the date on which you accessed the document.

A Note about Plagiarism

Research-based writing in Indian or international institutions, both educational and corporate, is filled with rules that writers, particularly beginners, are not

aware of or don't know how to follow. Many of these rules have to do with research and proper citation. Gaining familiarity with these rules, however, is critically important, as inadvertent mistakes can lead to charges of plagiarism - the unaccredited use (both intentional and unintentional) of somebody else's words or ideas. When using information from varied sources, you can avoid plagiarism by knowing what must be documented. As a rule of thumb, always provide reference and citation.

Specific words and phrases: When using an author's words verbatim, they must be placed in double quotation marks and due credit given to the source.

Information and Ideas: Even when paraphrasing or presenting someone else's ideas in your own words, you must document the source of the information.

Information: If a piece of information is not common knowledge, you need to provide a source, even if it is an unpublished document.

Ideas: As a writer, you must acknowledge the source/ author if it provided - points made and conclusions drawn, specific methods or theory, the arrangement of material, or a list of steps in a process or characteristics of a medical condition, etc.

Check Your Progress: 3

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this Unit.

1) List the different sub-sections of the final section of the research report.

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2) What are the different ways of documentation styles?

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3) Differentiate between conclusions and recommendations?

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15.6 EFFECTIVE WRITING

There are several aspects that makes writing effective. We have identified six features of effective writing.

Clear Objective: Having a clear objective is the most important feature of any writing. The purpose of all the sections and paragraphs in the report must be completely clear.

Good Organisation: A well-organised document always presents information in the Order of Importance. All sections should start with the conclusion and then elucidate it with supporting information.

Brief and Concise Writing: There should never be an information overload for the reader. It is prudent to be clear and concise and avoid unnecessary or peripheral information wherever possible. As a writer, you should be careful to use shorter sentences and familiar words.

Appropriate Language: You should avoid using too many jargons or technical words and should use simple words that may be easily understood by the readers. If you have to use a technical term, define it the first time you use it.

Relaxed Style: Style is something personal. It's 'how' you express your ideas rather than what you say. Your style should aim to be persuasive and interesting.

Correct Spelling, Grammar and Punctuation: Accuracy of the content is paramount. Although a good report may not directly depend on your language capabilities, it does depend on your ideas and ability to persuade others. Therefore, writers are advised to be accurate and to do a spell and grammar check.

Numbering: In this Unit, the point numbering system is used. Main sections are numbered 1, 2, 3 etc., with each sub-section given a decimal sub - division. This system of numbering is useful for readers to keep track of where they are within the report.

However, this is only one style of numbering and it is not necessary to always use it. Some documents, like, government reports may number all paragraphs consecutively and others may completely skip numbering. You, as a writer, should follow the technique that best fits in with your work.

Other Conventions

It is important as a writer for you to remember some other rules of presenting a report. The documents should ideally be typed, double-spaced on one side of the paper, with adequate margins; as an overcrowded page is difficult to read. You must also remember to number your pages and refer them correctly in the table of contents.

Most organisations have their own set of rules for presenting reports and other documents and these may vary from others. It is advisable that you follow the rules laid out by the organisation you work for.

Everyone has their own style of writing and this style should show through even when your material must be presented into a conventional format. Convey to your reader your enthusiasm for your investigation without using colloquial phrases (“Branded cumulative experiment on the roof”) or unnecessary scientific jargon (“most interesting investigation into the psychology of a member of the common Indian Diaspora that occurs on the geographic stratification of our local coastal Australian cities”). Write clearly and simply. Try reading it out aloud; if your sentences are such a mouthful that you stumble over them, redo them.

So who do you pitch the report at? How much information is necessary? Your friends, parents and a professional environmental scientist may read your report. The best idea is to write it for an intelligent person (for example, a colleague who is also doing communication research) who has a general knowledge of the subject like you. You do not need to go into detail, but you do explain new information or unusual methods that you alone know about.

Do whatever is necessary to include the readability. Do not hesitate to use the semicolon, short sentences, 15-20 words long, with one idea per sentence. Run on sentences can be very annoying; it is poor style to repeat a complex subject just to be exact. In some cases it may also be clearer to put paragraph indents, especially for separate mathematical formulae or separate quotations. Use active voice rather than passive voice.

Typing of report produces a document that looks good, but typing is not essential. There are a number of things to check (that are more visible on a typed report), whatever method you use. It is vitally important you do this. What sounded perfectly reasonable to you at 4 am in the morning often does not sound reasonable to anyone else on the planet the next day. By proofreading you also ensure that you are not making silly mistakes. Below is a checklist you should follow:

- 1) Typographical and spelling errors in the text
- 2) Mathematical or typographical errors in tables and graphs
- 3) Omissions of headings, and captions on tables and figures
- 4) Errors in the references
- 5) Try reading the work carefully (and slowly), sometime after you have finished.

Writing well is not easy. Even great authors find it difficult. The best way of writing is to write, edit, rewrite, and edit, revise, several times until the report is clear, concise and readable. One way of doing this is to have someone else read and/or edit your report.

Check Your Progress: 4

Note: 1) Use the space below for your answers.

- 2) Compare your answers with those given at the end of this Unit.

- 1) What are the formats of reference styles?

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2) What are the elements of editing and rewriting?

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3) What does 'Style' of writing mean?

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Activity -2

Download some research reports from the net or access them from the library and analyse the typical report format structures and variations in them.

15.7 LET US SUM UP

In this unit, we have taken you through the process of writing a research report in a simple and sequential manner. We discussed that a research report is a presentation of facts and findings, usually as a basis for recommendations; written for a specific readership, and probably intended to be kept as a record. A research report generally follows a typical format - the beginning, the main body of the report, and the final section. In fact, the structure of the unit, given at the beginning, can itself serve as a template for what should be included and how a report should be structured.

Writing a report is perhaps one of the most difficult parts of research process, because with so much material, it is hard to include everything. You are also so deeply involved that you sometimes do not see the errors. Hence due care needs to be taken while writing research report. In the final analysis, this report is what is presented either for evaluation by an examiner, for a panel of experts on the subject, and even for other researchers to read.

With this unit, we have come to the end of the course on Communication Research Methods. The purpose has been to provide you with enough inputs and knowledge so as to enable you to undertake systematic research in your area of interest and write the report. Remember, the more research you do, a better researcher you would become and contribute in generating scholarship in communication research. Practice makes perfect.

15.8 KEYWORDS

APA Format: APA refers to the American Psychological Association format for documentation. Typically, APA style is used for social science, biological science and nursing research papers.

Bibliography Card: A bibliography card contains the information needed when writing the Reference or Works Cited page. Use a different bibliography card for each reference source or keep the information in a file on your computer.

Bibliographic Information: The last page of your research paper or documented essay will list all of the sources you used to write your paper. Each source must contain bibliographic information including: author, title, publication information.

Citation: Each entry on the Works Cited or Reference page is called a citation, and it contains the bibliographic information needed for a reader to find the source information.

Journal: Journals are periodicals written by and for professionals in specific fields.

MLA Format: MLA refers to the Modern Language Association format for documentation. Typically, MLA style is used for language and literature research papers.

Paraphrase: One kind of citation to use is the paraphrase. When you paraphrase, you express another person's ideas in your own words. A paraphrase includes all of the information from the original source, but the wording is your own. All paraphrases must be documented with a parenthetical citation, and the source must be listed on the Works Cited or Reference Page of the paper.

Parenthetical Citation: Parenthetical citation is used to identify reference information in the paper. It is a brief notation placed at the end of the last sentence of each reference. It contains information needed to identify the source.

Periodical: Periodicals are publications that are printed at regular intervals, such as journals, magazines, newspapers, and newsletters.

Plagiarism: Plagiarism occurs when other people's words or ideas are used without giving credit to the source.

Primary Sources: Primary sources are the original source of information. They include original writings of an author, autobiographies, speeches, and laboratory experiments.

Quotation: A quotation is incorporating the exact wording of an author in your writing. Quotations should be used sparingly in your paper, and they must always be cited.

Reference Page: The reference page used with the APA format lists all of the sources used in compiling the paper. Each source must be fully documented with complete bibliographic information.

Secondary Sources: Secondary sources contain information about the primary source. They include newspapers, periodicals and journal articles written about a topic. Also, biographies, critiques and analyses are examples of secondary sources.

Summary: Summary is one kind of citation used in research papers. The summary captures the most important ideas from the original source. It is written entirely in your own words and is shorter than the original material. All summaries must be documented with a parenthetical citation, and the source must be listed on the Works Cited or Reference Page of the paper.

Works Cited Page: The Works Cited page used with the MLA format shows all of the sources used in compiling the paper or report. Each source is fully documented with complete bibliographic information.

15.9 REFERENCES AND FURTHER READINGS

John Bowden, *Writing a Report: How to prepare, write and present effective reports*, Publisher: How To Books, 2004

The Reporting Skills and Professional Writing Handbook, ELD Publications, 2008

http://wadsworth.cengage.com/english_d/special_features/plagiarism/WPAplagiarism.pdf

15.10 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress 1

- The three key sections of the report are the beginning, the main body of the report and the final section.
- The three sub-sections explain why it is interesting, what information is already known on the topic and what the aims of the study you have done were.

Check Your Progress 2

- The main body of the report contains many sections. It includes the survey of literature, the statement of the problem, the research design, the methodology, the data analysis, findings, and conclusion.
- A literature review summarises what has been researched and written about the subject by other researchers to date.
- Data can be presented in many ways. Quantitative data are presented in tables and graphs accompanied by descriptions and explanations. Qualitative data can be organised and presented by theme and with case studies and quotations from respondents among them.

Check Your Progress 3

- Summary, conclusions, recommendations, appendix, references.
- Both are ways in which citations are given in written reports.
- Conclusions state what the research has found in terms of previous theory or research; recommendations suggest a future course of action.

Check Your Progress 4

- Formats of reference style are APA, MLA, and Harvard.
- Clear objectives, good organisation, appropriate academic style, correct language and spelling.
- How an individual writes so as to convey thoughts to other people.